

NETWORK TRAFFIC ANALYSIS

Using Wireshark

SOC Investigation Report

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Tool Used	Wireshark, Command Prompt, Browser

1. Introduction

This report documents a hands-on network traffic analysis exercise carried out using Wireshark from a SOC analyst perspective. The goal was to capture real network traffic on a live machine and analyze each protocol to understand what normal traffic looks like, and more importantly, what suspicious traffic looks like.

As a SOC analyst, understanding how each protocol behaves in normal conditions is the foundation of being able to spot anything unusual. This exercise covered seven core protocols that appear in almost every network investigation.

2. Objective

The objective of this lab was to:

- Capture live network traffic using Wireshark on a Windows machine
- Apply protocol-specific filters to isolate and study each traffic type
- Analyze the packets and understand what each one means in plain terms
- Relate each protocol to real SOC investigation scenarios
- Document findings in a structured report format as a SOC analyst would

3. Tools Used

Tool	Purpose
Wireshark	Capturing and analyzing network packets in real time
Command Prompt (CMD)	Generating ICMP, ARP and DHCP traffic using ping, arp -a, ipconfig commands
Web Browser (Brave)	Generating DNS, TCP, TLS, HTTP and UDP traffic by visiting websites

4. Lab Sections

Each protocol was isolated using a specific Wireshark display filter. The analysis below explains what was observed in each capture and how a SOC analyst would use this information during a real investigation.

4.1 DNS - Domain Name System

The image shows a Wireshark network traffic capture of DNS traffic. The packet list pane displays 28 packets, all of which are DNS messages. The packet details pane shows the structure of a DNS Standard query response for b1.nel.goog, including the question section and the answer section with an A record (142.250.66.14) and a CNAME record (2404:6800:4007:804::200e). The packet bytes pane shows the raw data of the selected packet. The status bar at the bottom indicates that 2126 packets were displayed, with 28 (1.3%) shown and 0 (0.0%) dropped.

No.	Time	Source	Destination	Protocol	Leng	Info
308	10.900079900	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	100	Standard query 0x110c HTTPS beacons.gcp.gvt2.com
309	10.900286200	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	100	Standard query 0x010c AAAA beacons.gcp.gvt2.com
310	10.900416200	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	100	Standard query 0xdf60 A beacons.gcp.gvt2.com
312	10.939549400	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	213	Standard query response 0x110c HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com CNAME...
313	10.939551900	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	213	Standard query response 0x010c AAAA beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com CNAME...
314	10.939554800	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	172	Standard query response 0xdf60 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com CNAME gce...
1077	16.760985800	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	90	Standard query 0xe3d8 A google.com
1078	16.761212200	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	90	Standard query 0xe933 AAAA google.com
1079	16.814539500	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	106	Standard query response 0xe3d8 A google.com A 142.250.66.14
1080	16.815265700	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	118	Standard query response 0xe933 AAAA google.com AAAA 2404:6800:4007:804::200e
1099	19.134661600	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	92	Standard query 0xa7d1 HTTPS einthusan.tv
1100	19.135346400	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	92	Standard query 0x3512 AAAA einthusan.tv
1101	19.136031400	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	92	Standard query 0x42b5 A einthusan.tv
1105	19.140361200	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	164	Standard query response 0xa7d1 HTTPS einthusan.tv SOA master.rpz.airtelsam.com
1106	19.141287300	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	164	Standard query response 0x3512 AAAA einthusan.tv SOA master.rpz.airtelsam.com
1107	19.141288100	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	108	Standard query response 0x42b5 A einthusan.tv A 4.16.155.232
1607	19.767037100	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	98	Standard query 0xa30c HTTPS e2c34.gcp.gvt2.com
1609	19.768143800	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	98	Standard query 0x974d AAAA e2c34.gcp.gvt2.com
1610	19.772167000	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	98	Standard query 0x1dd3 A e2c34.gcp.gvt2.com
1612	19.799816100	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	114	Standard query response 0x1dd3 A e2c34.gcp.gvt2.com A 35.216.18.75
1613	19.802018300	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	155	Standard query response 0xa30c HTTPS e2c34.gcp.gvt2.com SOA ns1.google.com
1614	19.802020400	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	155	Standard query response 0x974d AAAA e2c34.gcp.gvt2.com SOA ns1.google.com
2079	25.051072900	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	91	Standard query 0xe73b HTTPS b1.nel.goog
2080	25.051442400	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	91	Standard query 0xbd48 AAAA b1.nel.goog
2081	25.051753500	fe80::1b99:ca8c:145...	fe80::ec28:d3ff:fee...	DNS	91	Standard query 0xccc8 A b1.nel.goog
2088	25.094323100	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	119	Standard query response 0xbd48 AAAA b1.nel.goog AAAA 2404:6800:4007:83f::2003
2089	25.095339300	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	151	Standard query response 0xe73b HTTPS b1.nel.goog SOA ns1.google.com
2090	25.095340200	fe80::ec28:d3ff:fee...	fe80::1b99:ca8c:145...	DNS	107	Standard query response 0xccc8 A b1.nel.goog A 74.125.200.94

Filter Used: dns

What it Shows

DNS traffic shows all domain name lookup requests made by the machine and the replies from the DNS server. Every time you visit a website, a DNS query happens first.

Analysis

The machine sent DNS queries to resolve multiple domains including google.com, beacons.gcp.gvt2.com, einthusan.tv, and b1.nel.goog. Key observations:

- The machine sent HTTPS, AAAA (IPv6), and A (IPv4) query types showing a complete modern DNS resolution flow
- CNAME records were returned for some domains showing alias chains before the final IP is given
- google.com resolved to IPv4 142.250.66.14 and IPv6 2404:6800:4007:804::200e confirming dual-stack resolution
- einthusan.tv returned a response from master.rpz.airtelsam.com meaning Airtel is blocking this domain using Response Policy Zone - a real DNS filtering technique
- Background service domains like gcp.gvt2.com and b1.nel.goog appeared

automatically showing the browser generates DNS traffic beyond what the user directly requests

SOC Use Case

SOC analysts look at DNS traffic closely because attackers frequently abuse it. Normal DNS looks exactly like what is captured here - known domains, clean responses, expected IPs.

- **Malicious domains** — checking if any queried domain matches known bad domains from threat intel feeds like VirusTotal or Cisco Talos.
- **Suspicious server responses** — if a DNS server returns an unexpected IP or routes through an unusual path like the Airtel RPZ block seen in this capture, it tells the analyst something about how the network is filtering or redirecting traffic
- **Random or very long domain names** like a1b2c3xyz.ru may indicate malware communicating with a C2 server using a Domain Generation Algorithm
- **DNS tunneling** — very high volume of DNS queries in a short time, especially with unusually long query names, can mean an attacker is secretly sending stolen data out of the network hidden inside DNS packets
- **Newly registered domains** being queried by internal machines are a strong indicator of phishing or malware activity since attackers register fresh domains specifically to avoid blacklists

4.2 ICMP - Internet Control Message Protocol

The image shows a Wireshark packet capture of ICMP Echo (ping) traffic. The filter used is 'icmp'. The capture shows a series of requests and replies between two hosts: 172.20.10.2 and 142.251.43.142. The packets are numbered 8 through 15 in the list, corresponding to sequence numbers 120/30720 through 123/31488. The protocol is ICMP, and the length is 74 bytes. The information column shows the sequence number, ttl, and the request/reply status.

No.	icmp	Source	Destination	Protocol	Leng	Info
8	icmpv6	172.20.10.2	142.251.43.142	ICMP	74	Echo (ping) request id=0x0001, seq=120/30720, ttl=128 (reply in 8)
9	icmpv6	142.251.43.142	172.20.10.2	ICMP	74	Echo (ping) reply id=0x0001, seq=120/30720, ttl=117 (request in 5)
10	icmpv6	172.20.10.2	142.251.43.142	ICMP	74	Echo (ping) request id=0x0001, seq=121/30976, ttl=128 (reply in 11)
11	icmpv6	142.251.43.142	172.20.10.2	ICMP	74	Echo (ping) reply id=0x0001, seq=121/30976, ttl=117 (request in 10)
12	icmpv6	172.20.10.2	142.251.43.142	ICMP	74	Echo (ping) request id=0x0001, seq=122/31232, ttl=128 (reply in 13)
13	icmpv6	142.251.43.142	172.20.10.2	ICMP	74	Echo (ping) reply id=0x0001, seq=122/31232, ttl=117 (request in 12)
14	icmpv6	172.20.10.2	142.251.43.142	ICMP	74	Echo (ping) request id=0x0001, seq=123/31488, ttl=128 (reply in 15)
15	icmpv6	142.251.43.142	172.20.10.2	ICMP	74	Echo (ping) reply id=0x0001, seq=123/31488, ttl=117 (request in 14)

The bottom of the image shows the packet details pane for the selected packet (No. 8), displaying the Internet Control Message Protocol (ICMP) header and data. The status bar at the bottom indicates 79 packets displayed, 8 (10.1%) displayed, and 0 (0.0%) dropped.

Filter Used: icmp

What it Shows	ICMP is used to test whether a device is reachable on the network. The most common use is the ping command which sends echo requests and waits for echo replies.
Analysis	The ping command was run targeting google.com. The machine at 172.20.10.2 sent 4 echo requests to Google at 142.251.43.142 and received 4 replies. Key observations: • 4 requests sent and 4 replies received - 100% connectivity with zero packet loss • TTL on outgoing requests is 128 (default Windows value), TTL on replies is 117 - the difference of 11 means the packet passed through 11 network hops to reach Google and return • Sequence numbers increment cleanly - 120, 121, 122, 123 - showing packets are sent and matched in order with no gaps • Each request is directly matched to its reply in the Info column confirming a clean request-response cycle with no dropped packets
SOC Use Case	ICMP by itself is harmless and is used every day for basic connectivity testing. But attackers misuse it in several ways that a SOC analyst must watch for. • Ping sweep - sending ICMP requests to every IP in a subnet to find which devices are online, this is a common first step in network reconnaissance • ICMP flood - sending thousands of pings to overwhelm a target, used in denial of service attacks • ICMP tunneling - hiding malicious data inside ICMP packets to communicate with C2 servers while bypassing firewall rules • Oversized or malformed ICMP packets are a red flag for packet crafting or old exploits like Ping of Death

4.3 TCP - Transmission Control Protocol

Wi-Fi

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tcp

No.	Time	Source	Destination	Protocol	Leng	Info
1	0.000000000	2401:4900:63fd:a005...	2a04:4e42:25::347	TCP	75	60533 → 443 [ACK] Seq=1 Ack=1 Win=510 Len=1
2	0.037586300	2a04:4e42:25::347	2401:4900:63fd:a005...	TCP	86	443 → 60533 [ACK] Seq=1 Ack=2 Win=270 Len=0 SLE=1 SRE=2
3	0.108944600	2401:4900:63fd:a005...	2404:6800:4003:c02:...	TCP	75	60233 → 5228 [ACK] Seq=1 Ack=1 Win=509 Len=1
4	0.243643300	2404:6800:4003:c02:...	2401:4900:63fd:a005...	TCP	86	5228 → 60233 [ACK] Seq=1 Ack=2 Win=1045 Len=0 SLE=1 SRE=2
7	8.641068100	74.125.68.94	172.20.10.2	TLSv1.2	127	Application Data
8	8.641709400	172.20.10.2	74.125.68.94	TCP	54	60635 → 443 [FIN, ACK] Seq=1 Ack=74 Win=514 Len=0
9	8.707033100	74.125.68.94	172.20.10.2	TCP	54	443 → 60635 [FIN, ACK] Seq=74 Ack=2 Win=1045 Len=0
10	8.707148600	172.20.10.2	74.125.68.94	TCP	54	60635 → 443 [ACK] Seq=2 Ack=75 Win=514 Len=0
11	11.596216200	172.20.10.2	74.125.68.94	TCP	55	60633 → 443 [ACK] Seq=1 Ack=1 Win=513 Len=1
12	11.596216200	172.20.10.2	74.125.68.94	TCP	55	60634 → 443 [ACK] Seq=1 Ack=1 Win=514 Len=1
13	11.694662400	74.125.68.94	172.20.10.2	TCP	66	443 → 60633 [ACK] Seq=1 Ack=2 Win=1041 Len=0 SLE=1 SRE=2
14	11.694665100	74.125.68.94	172.20.10.2	TCP	66	443 → 60634 [ACK] Seq=1 Ack=2 Win=1045 Len=0 SLE=1 SRE=2
15	12.379215400	2401:4900:63fd:a005...	2607:f8b0:4004:100b...	TCP	75	60652 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1
16	12.839444900	2607:f8b0:4004:100b...	2401:4900:63fd:a005...	TCP	86	443 → 60652 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SRE=2
17	12.957105600	172.20.10.2	54.187.119.242	TCP	55	60653 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1
18	13.248936700	54.187.119.242	172.20.10.2	TCP	66	443 → 60653 [ACK] Seq=1 Ack=2 Win=153 Len=0 SLE=1 SRE=2
20	14.234800100	2401:4900:63fd:a005...	2001:4860:4802:34:...	TCP	75	60655 → 443 [ACK] Seq=1 Ack=1 Win=508 Len=1
21	14.269581300	2401:4900:63fd:a005...	2001:4860:4802:34:...	TCP	75	60659 → 443 [ACK] Seq=1 Ack=1 Win=508 Len=1
22	14.269581700	2401:4900:63fd:a005...	2001:4860:4802:34:...	TCP	75	60657 → 443 [ACK] Seq=1 Ack=1 Win=508 Len=1
23	14.300144500	2001:4860:4802:34:...	2401:4900:63fd:a005...	TCP	86	443 → 60655 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SRE=2
24	14.312870000	2001:4860:4802:34:...	2401:4900:63fd:a005...	TCP	86	443 → 60657 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SRE=2
25	14.318058500	2001:4860:4802:34:...	2401:4900:63fd:a005...	TCP	86	443 → 60659 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
26	14.334502900	2401:4900:63fd:a005...	2001:4860:4802:34:...	TCP	75	60660 → 443 [ACK] Seq=1 Ack=1 Win=508 Len=1
27	14.334503100	2401:4900:63fd:a005...	2001:4860:4802:34:...	TCP	75	60661 → 443 [ACK] Seq=1 Ack=1 Win=508 Len=1
28	14.369471800	2401:4900:63fd:a005...	2001:4860:4802:34:...	TCP	75	60662 → 443 [ACK] Seq=1 Ack=1 Win=508 Len=1
29	14.379755600	2001:4860:4802:34:...	2401:4900:63fd:a005...	TCP	86	443 → 60660 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
30	14.383904300	2001:4860:4802:34:...	2401:4900:63fd:a005...	TCP	86	443 → 60661 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2

Frame 23: Reset, 86 bytes captured (808 bytes) on interface... 0000 18 1d ea 66 67 f5 ee 28 d3 e6 ea 64 86 dd 6b 8c ...fg...d.k

Transmission Control Protocol: Protocol

Packets: 46 · Displayed: 41 (89.1%) · Dropped: 0 (0.0%) | Profile: Default

Type here to search 29°C 16:15 24-06-2026

Filter Used: tcp	
What it Shows	TCP is the most widely used transport protocol. It provides reliable, ordered delivery of data and uses a three-way handshake to establish connections before any data is sent.
Analysis	The TCP filter showed active connections between the local machine at 172.20.10.2 and multiple Google servers. Key observations: - All traffic was going to and from port 443 confirming these are HTTPS encrypted web connections - ACK packets were consistently present showing data is actively being exchanged and acknowledged - FIN ACK packets were visible between 172.20.10.2 and 74.125.68.94 showing a clean and graceful connection termination where both sides agreed to close - A TLSv1.2 Application Data packet confirmed that the TCP connection is carrying encrypted data - SYN packets were not captured because the connections were already established before the capture started - this is completely normal - Both IPv4 and IPv6 connections were active at the same time showing dual-stack operation
SOC Use Case	TCP analysis helps SOC analysts understand how connections are being made. Most attack

traffic also uses TCP which makes it important to monitor closely.

- **SYN flood** - thousands of SYN packets sent without completing the handshake, used to exhaust server resources in a DDoS attack
- **Port scanning** - sequential SYN packets sent to many different ports from one source, indicating someone mapping what services are open
- Connections to unusual ports like 4444 or 1337 may indicate malware communicating with a C2 server
- Large data transfers over port 443 at unusual hours could be encrypted data exfiltration

4.4 HTTP - Hypertext Transfer Protocol

The image shows a Wireshark packet capture window with the filter 'http' applied. The packet list shows several HTTP requests and responses. The packet details pane shows the selected packet's structure, including the Hypertext Transfer Protocol section.

No.	Time	Source	Destination	Protocol	Leng	Info
1022	43.867736300	172.20.10.2	34.223.124.45	HTTP	568	GET /online HTTP/1.1
1029	44.207838300	34.223.124.45	172.20.10.2	HTTP	635	HTTP/1.1 301 Moved Permanently (text/html)
1063	47.754137300	2401:4900:63fd:a005...	2600:1f13:37c:1400:...	HTTP	588	GET /online HTTP/1.1
1068	48.026520600	2600:1f13:37c:1400:...	2401:4900:63fd:a005...	HTTP	119	HTTP/1.1 301 Moved Permanently (text/html)
1073	48.040903300	2401:4900:63fd:a005...	2600:1f13:37c:1400:...	HTTP	589	GET /online/ HTTP/1.1
1086	48.343916000	2600:1f13:37c:1400:...	2401:4900:63fd:a005...	HTTP	521	HTTP/1.1 200 OK (text/html)
1090	48.494411200	2401:4900:63fd:a005...	2600:1f13:37c:1400:...	HTTP	497	GET /favicon.ico HTTP/1.1
1091	48.856226100	2600:1f13:37c:1400:...	2401:4900:63fd:a005...	HTTP	490	HTTP/1.1 200 OK (PNG)

Filter Used: http

What it Shows

HTTP is the protocol used for web communication. Unlike HTTPS it sends everything in plain text with no encryption, meaning anyone on the same network can read the full content of requests and responses.

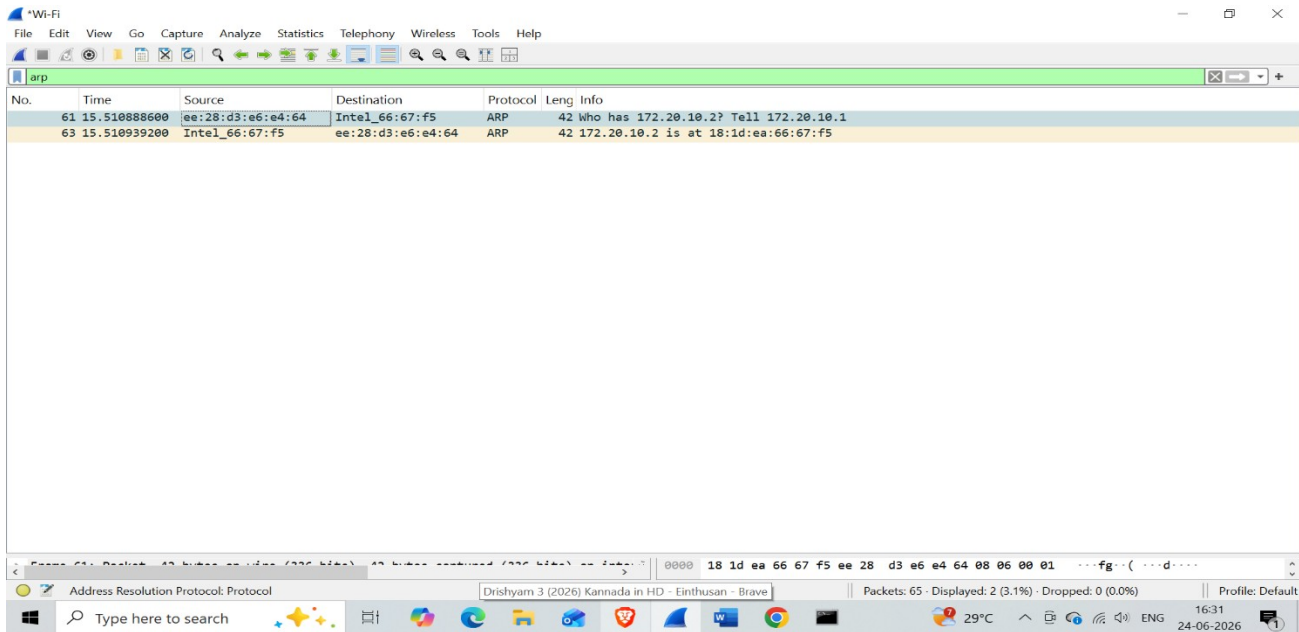
Analysis

The browser was directed to an HTTP website to generate plain HTTP traffic. The machine at 172.20.10.2 communicated with a web server at 34.223.124.45.

Key observations:

	• The browser sent an HTTP GET /online request asking the server to send back a webpage • The server replied with HTTP 301 Moved Permanently redirecting the browser to the HTTPS version of the site as a security measure • After the redirect the server returned HTTP 200 OK with a text/html response confirming the page loaded successfully • The browser also made a GET /favicon.ico request to fetch the small icon shown in the browser tab, which returned 200 OK with a PNG image • All traffic is HTTP/1.1 and completely unencrypted - everything is readable in plain text by anyone capturing this traffic on the same network
SOC Use Case	HTTP is dangerous from a security perspective because of its lack of encryption. SOC analysts look at HTTP traffic for signs of both data theft and active attacks. • Credentials in plain text — if a user submits a login form over HTTP instead of HTTPS, the username and password are fully visible in the packet. An attacker on the same network can steal them without doing anything complex • Suspicious request paths — GET or POST requests to paths like /shell, /cmd.php, /upload, or /admin in HTTP traffic may indicate someone is accessing a web shell, exploiting a vulnerability, or trying to upload a malicious file to the server • HTTP on non-standard ports — legitimate web traffic uses port 80. If an internal machine is making HTTP requests to ports like 8080, 8888, or other random high ports it could be malware communicating with a C2 server disguised as normal web traffic • Downgrade attack — a machine that keeps using HTTP even after receiving a 301 redirect to HTTPS may have been intercepted by an attacker who is stripping the encryption, forcing the victim to stay on plain HTTP so the attacker can read everything

4.5 ARP - Address Resolution Protocol



Filter Used: arp

What it Shows

ARP is used inside a local network to find out which MAC address belongs to a given IP address. Before any device can send data to another device on the same network it needs to know the MAC address, and ARP is how it finds that.

Analysis

Only 2 ARP packets were captured showing a clean and complete ARP exchange. The router needed to find the MAC address of the local machine. Key observations:

- Packet 61 shows an ARP Request from the router MAC ee:28:d3:e6:e4:64 asking Who has 172.20.10.2? Tell 172.20.10.1 - the router was looking for the local machine's MAC address before it could deliver traffic
- Packet 63 shows the ARP Reply from the local machine's Intel adapter 18:1d:ea:66:67:f5 confirming 172.20.10.2 is at 18:1d:ea:66:67:f5
- Both packets completed in under 1 millisecond showing an instant and clean ARP resolution
- Only 2 packets captured which is the most minimal possible ARP exchange - one request and one reply with no retransmissions or errors

SOC Use Case

ARP has no authentication built in. Any device on the network can send a fake ARP reply and other devices will blindly accept it. This makes ARP a common target for local network attacks.

- **ARP Spoofing or ARP Poisoning** - an attacker sends fake ARP replies claiming their MAC address belongs to the gateway IP. All traffic then flows through the attacker enabling a Man in the Middle attack
- **Gratuitous ARP** - an unsolicited ARP reply sent without a request, often used to overwrite ARP entries on other devices

- ## 4.6 UDP - User Datagram Protocol

- QUIC protocol packets were visible from packet 61 - QUIC is Google's modern protocol that runs over UDP and is used by YouTube and Chrome. The initial handshake showed CRYPTO, PADDING, and PING packets followed by ACK confirmations
- Port 56807 was the local machine's temporary source port assigned for this specific session
- 980 out of 1035 total packets (94.7%) were UDP showing how dominant UDP traffic is during normal browser activity

SOC Use Case

UDP is fast and hard to track because there is no connection state. Attackers take advantage of this in several ways.

- **DNS tunneling** - attackers encode stolen data inside DNS query names and send it out over UDP port 53. Unusually long domain names or very high DNS query volumes are the main signs
- **UDP flood DDoS** - sending large volumes of UDP packets to random ports on a target to exhaust its bandwidth
- **QUIC abuse** - because QUIC is encrypted and uses UDP it is harder for security tools to inspect, making it useful for hiding C2 traffic
- UDP traffic on unexpected ports from internal machines may indicate malware activity or data exfiltration

4.7 DHCP - Dynamic Host Configuration Protocol

Wireshark packet capture showing DHCP traffic. The interface is 'Wi-Fi'. The filter is 'dhcp'. Two packets are visible:

No.	Time	Source	Destination	Protocol	Leng	Info
18	6.414018100	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xbdd2e6c2
21	6.544182900	172.20.10.1	172.20.10.2	DHCP	346	DHCP ACK - Transaction ID 0xbdd2e6c2

The DHCP ACK packet details show 'Transaction ID 0xbdd2e6c2'.

Filter Used: dhcp (bootp)	
What it Shows	DHCP is the protocol that automatically assigns an IP address to a device when it connects to a network. It follows a four step process called DORA - Discover, Offer, Request, Acknowledge.
Analysis	The ipconfig /release and /renew commands were used to force a DHCP exchange. The machine requested to keep its previous IP address. Key observations: • Packet 18 shows a DHCP Request sent from source 0.0.0.0 to broadcast 255.255.255.255 - the machine used 0.0.0.0 because it had no confirmed IP yet and broadcast because it did not know the router address • The machine was asking the router - Can I keep using my previous IP address 172.20.10.2? • Packet 21 shows a DHCP ACK from the router at 172.20.10.1 to the machine at 172.20.10.2 confirming - Yes, that IP is yours • Both packets share the same Transaction ID 0xbdd2e6c2 which links the request and reply together • Only 2 packets instead of the full 4 step DORA process because the machine already knew its previous IP and jumped straight to Request and Acknowledge - normal behaviour when reconnecting to a known network
SOC Use Case	DHCP has no authentication by default which means any device on the network can pretend to be a DHCP server. This makes it a target for both attackers and network misconfigurations. • Rogue DHCP server - an attacker sets up their own DHCP server and responds faster than the real router, assigning themselves as the gateway. All traffic then routes through the attacker enabling a Man in the Middle attack • DHCP starvation - an attacker sends thousands of DHCP requests with fake MAC addresses to use up all available IP addresses, kicking legitimate devices off the network • Unauthorised device detection - DHCP logs show every device that joined the network with its MAC address and timestamp. Unknown devices stand out immediately in the SIEM • IP conflict detection - two DHCP replies assigning the same IP to different MACs indicates a rogue server or misconfiguration

5. Investigation Summary

When you open a website, multiple protocols work together in a specific order. Here is the full flow from start to finish:

#	Protocol	What happens
1	DHCP	When your machine connects to the network, DHCP assigns it an IP address automatically
2	ARP	Your machine uses ARP to find the MAC address of the router so it can send traffic out
3	DNS	When you type a website name, DNS converts it to an IP address your machine can connect to
4	TCP	Your machine does a three-way handshake with the web server to establish a reliable connection
5	TLS	Over that TCP connection, TLS negotiates encryption so all communication is secure
6	HTTP/HTTPS	The actual web page data is transferred over the encrypted connection
7	ICMP	Used separately to test connectivity - ping checks if a host is reachable
8	UDP	Used by DNS, video streaming via QUIC, and any real-time data that needs speed over reliability

6. Conclusion

This project gave me a clear, hands-on understanding of how different network protocols behave in real traffic. By capturing live packets and applying filters one by one, I was able to see exactly what normal DNS lookups, ICMP pings, TCP connections, HTTP requests, ARP exchanges, UDP streams, and DHCP assignments look like from the inside.

More importantly, this exercise helped me understand what each of these protocols looks like when something is wrong. A SOC analyst spends a lot of time in tools like Wireshark looking for the things that do not fit - unusual domains, unexpected ports, abnormal packet volumes, or suspicious connection patterns.

Having this foundation of protocol knowledge makes it much easier to spot those anomalies quickly and confidently during a real investigation. This is directly applicable to day-to-day SOC work including alert triage, incident investigation, and threat hunting.

--- End of Report ---

